

Flood Susceptibility Literature Review

14 papers, classified by function · compiled from Literature Review og / (2) / (3).xlsx and the accompanying PDF · 9 September 2026

SOURCE RECONCILIATION.

The four files in the folder are one review table split across formats. The PDF is a single-page print of the first four rows of og.xlsx; the other two spreadsheets carry additional rows. Deduplicated, the set contains **14 distinct papers**. One correction: in og.xlsx row 3 the “Authors & Year” cell was overwritten with the stray text data uncertainty — the PDF preserves the correct value, Antzoulatos et al. (2022), Sustainability.

1 Event-time flood detection & hydraulic modelling

Takes an event as input — a storm, a discharge — and returns observed or simulated flood extent and depth. *Event-dependent by construction: no event, no output.*

PAPER	ALGORITHM	HEADLINE RESULT	PRINCIPAL LIMITATION
Integrating deep learning, satellite image processing and spatial-temporal analysis for urban flood prediction Mohamadiazar, Ebrahimian & Hosseiny (2024), J. Hydrology · Miami-Dade, FL BASE PAPER	CNN (U-Net) on Sentinel-1 SAR	F1 89.85%, OA 93.73%, AUC 0.93; Ground Truth Index 84.05%	Requires live same-day rainfall; cannot assess an empty plot outside a storm; no per-factor explainability
Flood Hazard and Risk Mapping by Applying an Explainable ML Framework Using Satellite Imagery and GIS Data Antzoulatos et al. (2022), Sustainability · Trieste / Muggia / Monfalcone, Italy	Random Forest (best), SVM, Naïve Bayes, NN	P/R/F1 $\approx$ 0.99 all hazard classes; depth + velocity = 66% of importance	Labels need a calibrated 2D hydraulic model, unavailable for most basins including the Periyar
Urban flood susceptibility mapping using remote sensing, social sensing and an ensemble ML model Zhu, Guo & Huang (2024), Sustainable Cities and Society	XGBoost, SVC, MLP, MDL $\rightarrow$ RF ensemble	Ensemble accuracy 0.942 train / 0.940 test, beating every standalone model	Multimodal + ensemble stages are markedly more complex than plain RF or XGBoost
Unravelling the complexities of urban fluvial flood hydraulics through AI Al Mehedi, Smith, Hosseiny & Jiao (2022), Scientific Reports · Lower Darby Creek, PA PRIOR ART — UICA	LR / DT / SVM / KNN classify; DNN regresses depth	DT F1 0.991, LR Jaccard 0.995; DNN depth RMSE 0.027	Surrogate for a calibrated hydraulic model, so still needs one; discharge is a live input; impervious feature is static

2 Static susceptibility mapping

No event input. Terrain, hydrology and land-cover covariates yield a standing susceptibility score for every pixel, plus an account of who is exposed to it. *These are the papers your baseline must match.*

PAPER	ALGORITHM	HEADLINE RESULT	PRINCIPAL LIMITATION
High-resolution flood susceptibility mapping and exposure assessment in Pakistan Waleed & Sajjad (2025), IJDRR · national scale, FABDEM 30 m NOVELTY HOOK	LightGBM & XGBoost + SHAP	F1 0.86, AUC 0.81 (10-fold); 29% of Pakistan critical; 94.7 M people exposed	Self-stated: does <i>not</i> integrate land-use change as a driver — named explicitly as future work
Multi-criteria decision analysis for regional-scale flood susceptibility mapping in Kerala state, India Kendagannaswamy, Roopa, Harish & Mukesh (2025), Discover Applied Sciences GEOGRAPHIC BENCHMARK	LightGBM + AHP (dual model)	LightGBM AUC 0.948 vs AHP 0.769; 18.5% of Kerala very vulnerable	Self-stated: conventional random split, no spatial or temporal CV; SRTM canopy bias; single static snapshot
A Systematic Review of Urban Flood Susceptibility Mapping Islam, Zeleke, Afroz & Melesse (2025), Remote Sensing · 120 studies, 2012–2024 REVIEW	Comparative survey: MCDA, FR, LR, SVM, RF, ANN, CNN	SVM 82–96% across studies; ML consistently beats expert-weighted MCDA	Contributes no benchmark of its own; stops Sept 2024; does not treat validation design as a first-class issue

3 Scenario & change-driven susceptibility

The distinguishing move: train on the present, re-score the trained model on a modified land-use layer, then difference the two maps. *This is the operation your project performs at parcel scale.*

PAPER	ALGORITHM	HEADLINE RESULT	PRINCIPAL LIMITATION
Future flood susceptibility mapping under climate and land use change Khodaei, Nasiri Saleh, Nobakht Dalir & Zarei (2025), Scientific Reports · Kashkan, Iran CLOSEST TEMPLATE	RF, XGBoost, SVC — each PSO-tuned; CA-Markov + CMIP6	RF AUC 0.98; high-risk zone 20%; under SSP585 high-risk +36.26 km ²	Random negative sampling and a plain 70/30 split; basin rather than parcel scale; LULC bundled with climate, so the pure development effect is never isolated
Investigation and Evaluation of LULC Change Effects on Current and Future Flood Susceptibility Rashidiyan & Rahimzadegan (2024), Natural Hazards Review	ANN, LR, RF, SVM + Land Change Modeler	SVM best: RMSE 0.125, R ² 0.9373	Future-LULC simulation adds a second modelling layer on top of the flood model itself
Assessing the impact of climate and land use change on flood vulnerability, coastal Tamil Nadu Abijith, Saravanan, Parthasarathy et al. (2025), Geoscience Letters	Random Forest; LULC projected to 2030 / 2040 / 2050	Moderate and very-high risk areas grow across every future projection	Couples future climate scenarios to LULC projection, making the pipeline broader than a minimum viable study

4

Methodology — labels, sampling and honest validation

These produce no map at all. Their subject is whether anyone else's numbers can be trusted. *Three of the four exist specifically to show that a headline AUC from a random split means very little.*

PAPER	ALGORITHM	HEADLINE RESULT	PRINCIPAL LIMITATION
Data Uncertainty of Flood Susceptibility Using Non-Flood Samples Zhang, Wei, Yao, Sun, Zhen & Xia (2025), Remote Sensing · Zijiang River Basin, China LOAD-BEARING	FR + One-Class SVM negative screening; RF, AdaBoost, GBDT, stacking; SHAP	Ensemble AUC 0.71 / 0.72 / 0.94 / 0.95 across four negative sets — a 24-point swing from sampling alone; naïve negatives scored a median susceptibility of 0.87	One rural basin (built-up only 2.1%), so untested in dense urban settings; its own 0.95 still comes from a random split
A Positive-Unlabeled Learning Algorithm for Urban Flood Susceptibility Modeling Li, Liu, Liu, Gao, Huang & Huang (2022), Land · Guangzhou, China PRIOR ART — PU	PBLC (positive + background with constraints), benchmarked against ANN	AUC $\approx$ 0.90 for both, but PBLC far better on F_{pb} ; label constant $c = 0.58$, calibrated AUC $\approx$ 0.96	Could not test PBLC with a CNN (only 532 positives); omits soil, geology and drainage density for want of data
Spatio-temporal cross-validation to predict pluvial flood events in the Metropolitan City of Venice Zanetti, Allegri, Sperotto, Torresan & Critto (2022), J. Hydrology LEAKAGE EVIDENCE	LR, NN, RF under Random CV vs Leave-Location-and-Time-Out CV	Random CV is over-optimistic because nearby locations and repeated events correlate; LR under LLTO-CV most reliable on unseen space-time	Models pluvial flood <i>events</i> rather than a standing vulnerability score
Challenges in data-driven geospatial modeling for environmental research and practice Koldasbayeva, Tregubova, Gasanov, Zaytsev, Petrovskaia & Burnaev (2024), Nature Communications GENERAL PRINCIPLE	Algorithm-agnostic; the subject is validation design, imbalance, error structure and uncertainty	Documents cases where ignoring spatial distribution produced deceptively high predictive power; proper spatial validation exposed predictors with poor target relationships	Domain-general rather than flood-specific; underwrites the protocol, not the results

How the classification maps onto the project

1 • Event-time Supplies the flood inventory : Sentinel-1 → Otsu → binary flood extent, and the evaluation metrics used for comparability.	2 • Static Supplies the baseline and the benchmark to beat — the Kerala AHP + LightGBM pair, re-run under honest validation.	3 • Scenario Supplies the ΔS operation : train once, re-score on a developed land-use layer, difference the maps, report km ² per class.	4 • Methodology Supplies the evaluation protocol — PU learning, screened negatives, spatial block CV — that makes ΔS defensible.
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The four categories align almost exactly with the four stages set out in the “Novelty for Our Project” column of the source table. Categories 1 to 3 are what the work builds on; category 4 is what stops the build from being an artefact of sampling.

Cross-cutting observations

- **Two papers deliberately straddle categories, and both are credit-where-due citations.** Al Mehedi et al. (cat. 1) already published a static version of the upslope-impervious-contributing-area feature, and Li et al. (cat. 4) already published positive-unlabeled learning for urban flood susceptibility. Neither can be claimed as novel; both should be cited and differentiated.
- **The labelling problem is what separates cat. 1 from cat. 2 in practice.** Antzoulatos et al. and Al Mehedi et al. both derive labels from a calibrated 2D hydraulic model. No such model exists for the Periyar basin, which is precisely why satellite-observed flood extent plus PU learning is the route available to you.
- **Every reported accuracy in categories 1 to 3 rests on a random train/test split.** Kendagannaswamy et al. and Khodaei et al. both name this in their own limitations. Category 4 quantifies what it costs: a 24-point AUC swing from negative sampling alone (Zhang et al.) and demonstrable over-optimism from spatial correlation (Zanetti et al.).
- **Two papers name your contribution as their own future work.** Waleed & Sajjad state that integrating land-use and urban-sprawl change into susceptibility remains undone; Islam et al. call for LULC projection scenarios in urban susceptibility mapping. The gap can therefore be cited rather than asserted.
- **Ranking metrics conceal the failure mode that matters here.** Li et al. show AUC stays flat at 0.90 while calibration diverges sharply — an argument for reporting PR-AUC and Brier score alongside AUC when the deliverable is a *difference* between two susceptibility surfaces rather than a ranking.

Classification derived from the Datasets, ML Algorithm, Methodology, Results and Contribution/Limitation columns of the source review table. Category assignment follows what each paper *does* — its input requirement and its output object — rather than the algorithm it happens to use.